UNIVERSITY EXAMINATIONS

28 June 2026

COS1501

Theoretical Computer Science

25 marks

Duration : 1 Hours

Welcome to the COS1501 Assignment 3.

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This paper consists of 3 pages.

Instructions:

- Please note that NO other form of assistance is allowed (another individual, notes, internet or anything else).
- Cutting and pasting from the textbook (electronic media) is prohibited.
- You have 1 hours to complete this paper.
- You may upload PDF files only, nothing else is acceptable.
- Be sure that your upload is NOT password protected or uploaded as read only.
- Answer all questions and in sequence.

- Incorrect file format and uncollated answer scripts will not be considered.
- NO emailed scripts will be accepted.
- Students are advised to preview submissions (answer scripts) to ensure legibility and that the correct answer script file has been uploaded.
- Incorrect answer scripts and/or submissions made on unofficial assessment platforms (including the invigilator cell phone application) will not be marked and no opportunity will be granted for resubmission.
- Submissions will only be accepted from registered student accounts.
- Students suspected of dishonest conduct during the assessment will be subjected to disciplinary processes. UNISA has a zero tolerance for plagiarism and/or any other forms of academic dishonesty.

Question 1 [Theory and Understanding]

1.1 Explain the following concepts in your own words. Your explanations should reflect the underlying mathematical principles and use appropriate notation where necessary.

a. Set Relation (1)
b. Function (1)
c. Reductio ad absurdum (1)
d. Domain (1)
e. Trichotomy (1)

Sub Total [5]

Question 2 [Concept and application]

2.1 Let Q = {(1,a), (3,b), 5, (7,c), (9,d), 11} and T = {2,4} be sets. Find the Cartesian product of the following Q x T, and describe briefly what your steps (2)

2.2 Let S = {(e,e), (e,p), (p,e), (q,r), (p,s)} be a relation on the set H = {e,p,q,r,s}.

a. Examine the relation S and state with justification, whether it is reflexive, irreflexive, symmetric, antisymmetric, and transitive (5)

2.3 Let P be defined as the inverse relation of S.

a. Determine the relation composition P;S (3)

Sub Total [10]

Question 3 [Complex Theorems and Problem Solving]

3.1 Proof that if R is a relation on a set of Positive Integers $\mathbb{Z}$, then R is symmetric if and only if R is the same as its inverse. (2)

3.2 Let W = {1,2,3} be a set, and G = {, {1}, {2,3}} be defined from W. Determine if G is a partition from W, and if it is, describe it's equivalence relation. (3)

3.3 Let D = {(x,y) | y = x+1} be a relation from X to Y, where X=Y=$\mathbb{Z}$. Determine if D is a function (2)

3.4 Let f:$\mathbb{Z}$→$\mathbb{Z}$ be a function defined by f(x) = 3-x. Check if the function comply with the property f injectivity, surjectivity, and bijectivity (3)

Sub Total [10]

TOTAL [25]